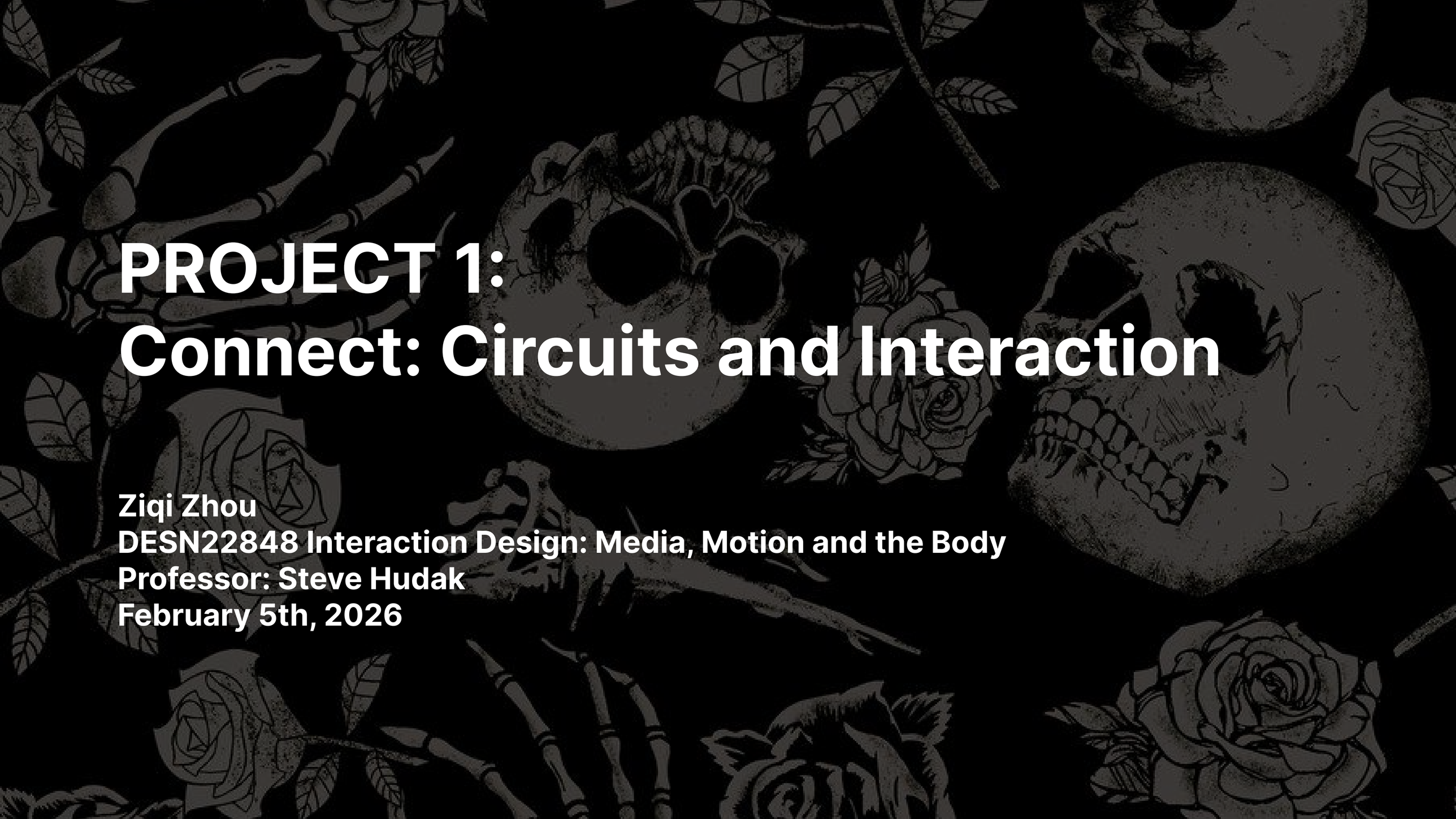


PROJECT 1:
Connect: Circuits and Interaction

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DESN22848 Interaction Design: Media, Motion and the Body
Professor: Steve Hudak
February 5th, 2026

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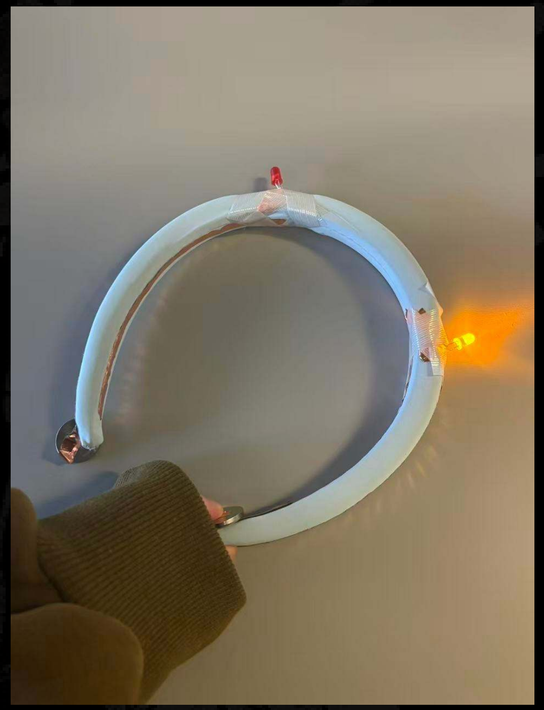
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Design Statement: Wearable





Project Name:
Glow Headband

Concept:
This is a portable, wearable interactive device designed for temporary nighttime illumination. It provides instant lighting in darkness or hands-free illumination when both hands are occupied.
Its core structure consists of conductive copper foil, a button battery, and an LED light.
My design concept involves activating the circuit through external pressure on the internal battery to control the headband's illumination—turning it on and off. The final design closely aligns with my initial vision.

Design Principles:
- Immediate Response:

Enables instant light control through simple pressure actions, delivering direct and clear interactive feedback. (e.g., placing the headband on the head activates the circuit via external pressure on the internal battery.)

- Contextual Integration:

Balances decorative appeal as a hair accessory with practicality for specific scenarios (e.g., nighttime activities, parties).

- Lightweight & Comfortable:

Utilizes a simplified circuit and soft external materials to ensure wearing comfort without compromising the headband's functionality. (The headband cannot be disassembled to showcase circuit materials. However, installing the circuit internally would enhance usability.)

Interaction:
When worn, pressure from the headband causes its tail section to press against the back of the ear. Pressing the battery area at the tail connects the battery to the copper foil circuit, activating the LED lights. (Releasing the pressure disconnects the circuit, extinguishing the lights.)

Suitable For:
Ideal for nighttime outdoor activities, concerts, parties, themed photo shoots, or any occasion requiring ambient lighting and playful decoration.

Design Statement: Non-Wearable





Project Name:
Temporary Night Light

Concept:
This multifunctional temporary night light serves as a convenient, space-saving reading light while doubling as a bookmark and refrigerator magnet. Crafted from copper foil, batteries, an LED light, a flat silicone figurine, and magnets. My design concept involves activating the circuit by applying pressure to the external battery, thereby controlling the headband's illumination. However, due to the object's small size, it can serve as a multifunctional tool and decorative piece.

Design Principles:
- Multifunctional:

Integrates lighting, bookmark, and decorative functions into one item, enhancing usability and appeal.

- Intuitive Feedback:

Place the item in the designated position and apply pressure to the external battery to activate the circuit, controlling the headband's illumination and deactivation.

- Flexible & Portable:

Utilizing magnetic adhesion, it can be easily attached to books, refrigerators, or metal bedheads, adapting to various usage scenarios.

Interaction:
Pressing the external battery activates the circuit, connecting the battery to the copper foil circuit to illuminate the LED light. Releasing pressure disconnects the circuit, causing the light to turn off.

Suitable For:
Nighttime readers, students, dormitory living, as a cozy mini mood light, or any scenario requiring temporary, localized illumination.

Design Statement: Wearable | Course Theory & Design Foundation

This project builds upon the concept of “Physical Computing” introduced in Week 1, utilizing simple circuits to bridge the physical world with digital information.

In theoretical application, my design deeply integrates key frameworks from the previous weeks of the course:

1. Affordance and Intuitive Interaction

My design directly applies and validates Week 1's principle: “An action possibility in the relationship between a user and an object” (Week 1, Slide 63). In the Glow Headband, the headband's wearability and elasticity, combined with the battery's compressibility, collectively “suggest” an interaction activated by wearing and pressing. The conductivity and ductility of copper foil “permit” circuit construction on flexible fabrics. The design eliminates redundant buttons or instructions, instead allowing the material's physical properties to guide users toward naturally discovering interactions—transforming abstract theory into intuitive, tactile design principles.

2. The Body as an Interface and Interruptive Interaction

Week 3's discussion on “The Body as an Interactive Medium” profoundly inspired my wearable design. Selecting a headband as the carrier leverages its intimate contact with the body (head), making interaction a natural extension of bodily posture. For the interaction mechanism, I adopted a clear “Interruptive Switch” logic. In the default state, the battery is not fully in contact with the circuit, creating a physical “interruption” that keeps the light off; When the user applies pressure, the bodily action (finger pressure) pushes the battery into its designated position, “completing” the circuit and illuminating the light. This “interrupt-complete” mechanism directly and unambiguously translates bodily action into system response, perfectly embodying the course's emphasis on “Body Action Triggers Response.”





Project Name: Glow Headband

Design Statement: Wearable | Material, Technology, Operation

- **Material:**

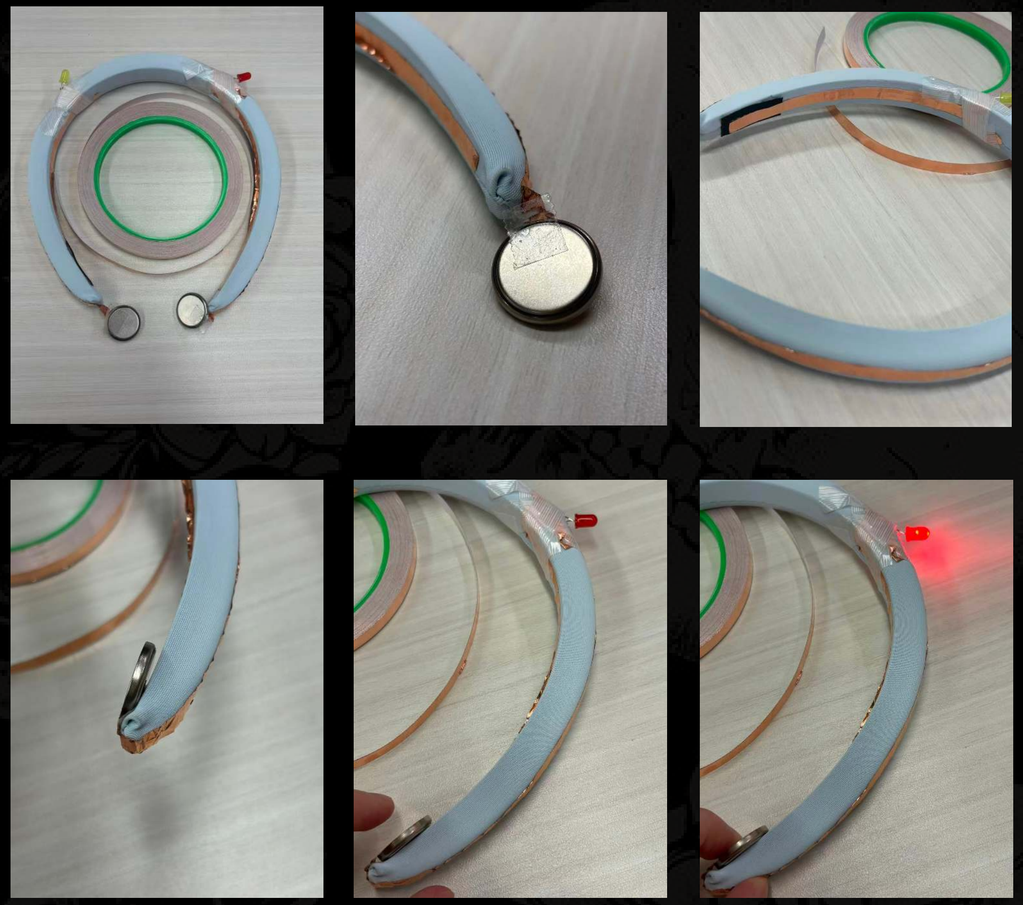
Conductive Copper Foil Tape: Utilized as the primary circuit conductor on the headband due to its flexibility and excellent conductivity, it conforms well to fabric surfaces.
Battery: Serves as a low-voltage power source meeting safety design requirements, while also functioning as a movable switch component.
LED Light: Selected for its compact size and low power consumption, facilitating integration into the headband. Its brightness is moderate, avoiding excessive glare.

- **Technology:**

Circuit Design: Utilizes a simple series circuit. The battery's positive and negative terminals connect to copper foil contacts via pressure, forming a complete current loop.
Switch Mechanism: The battery itself serves as the sole switch. Physically pressing changes the battery's position, connecting or breaking the circuit. Interaction is simple with immediate, visible results.
Structural Layout: The circuit is cleverly positioned inside or on the side of the headband. (However, concealing or encapsulating all materials internally ensures safe wear and aesthetic appeal.)

- **Operation:**

Press to illuminate: The user presses the battery at a specific location on the headband with their hand or head. This contact between the battery and the copper foil circuit below causes the LED light to instantly illuminate.
Release to extinguish: When the user stops pressing, the battery slightly springs back due to the internal elastic material, disconnecting from the circuit and causing the light to turn off.
Interaction Loop: The entire operation flows naturally and intuitively, aligning with the “press to light, release to extinguish” cognitive expectation, requiring zero learning curve.



Design Statement: Wearable | Design & Visual Inspiration

This design draws inspiration from reimagining the everyday object of the “headband.” We explore its potential not merely as a tool for securing hair, but as a wearable interactive medium.

Visually and culturally, I drew inspiration from:

- **Music festivals and carnival culture:**

In these settings, luminous accessories serve as essential props for expressing individuality and immersing oneself in the atmosphere. The light-up headband inherits this tradition of “decorative light effects,” transforming it into a personally controllable interactive experience.

- **Science fiction and futuristic aesthetics:**

Characters in many sci-fi works often wear gear with luminous elements, symbolizing technological sophistication or special abilities. This light-up headband can be seen as a more accessible, playful interpretation of this aesthetic, bringing futuristic imagination into everyday entertainment.

Regarding the light effects design, I initially intended to use a uniform color temperature (such as warm white or cool blue) to avoid clutter and create a harmonious atmosphere. However, I discovered that my materials lacked LEDs in the same color family, forcing me to use different light colors. If the light colors were consistent, it would have balanced the decorative and functional aspects.















Design Statement: Wearable | Work & Iteration

Conceptualization and Prototyping Phase:

I initially conceived two circuit switch solutions:

Solution A (Magnetic Switch):

- Small magnets at both ends of the headband would attract each other, applying pressure to activate hidden contacts. However, circuit stability was highly dependent on magnetic force and alignment precision, leading to its rejection.
- Option B (Pressure Switch): The ultimately adopted solution. Direct external pressure on the internal battery connects the circuit. Advantages include simple, reliable structure and extremely direct, forceful interactive feedback. After testing with a basic model, I abandoned Option A due to its low success rate. Though Option B seemed simpler and more “brute-force,” it ensured a clear light effect response with every press, better meeting the certainty required for “entertainment interaction.”





Testing and Iteration Phase:

During initial physical prototype testing, two primary issues emerged:

- Unstable battery contact:

Relying solely on pressure caused the battery to shift, resulting in flickering lights.

- Solution:

Added a small piece of double-sided tape behind the battery. This secures the battery in place while providing elastic pressure, ensuring the circuit disconnects when released. It also increases contact area and stability during pressing.

- LED detachment:

The headband's inherent flexibility caused frequent bending, leading to LED edges lifting and detaching.

- Solution:

After securing the LEDs with copper foil, I covered and encapsulated the entire circuit area using transparent flexible fabric tape (e.g., cloth tape). This significantly enhanced circuit durability without compromising appearance or tactile feel.





Design Statement: Non-Wearable | Course Theory & Design Foundation

This project builds upon the concept of “Physical Computing” introduced in Week 1, utilizing simple circuits to bridge the physical world with digital information.

In theoretical application, my design deeply integrates key frameworks from the previous weeks of the course:

1. The Practice Philosophy of Low-Fidelity Prototyping

The exploration of “the essence of prototyping” in Week 3 (Slides 47-55) liberated my design process. In Activity 1, experiments using basic materials like graphite and copper tape made me deeply appreciate that materials themselves are “symbols” carrying function and meaning. This prompted me to steadfastly use the simplest, most readily available materials—copper foil, batteries, LEDs—for rapid iteration throughout the project. I prioritized validating the reliability of interaction logic and the stability of circuit functionality over prematurely fixing the appearance. This “low-fidelity” philosophy enabled me to test core concepts agilely. In designs like the “Temporary Night Light,” I explored the potential of integrating circuits with various everyday objects (bookmarks, silicone dolls, magnets), ensuring an open design mindset and prioritizing functionality.

2. Contextual Implicit Interaction in Non-Wearable Devices

For non-wearable devices like the “Temporary Night Light,” my design emphasizes contextual implicit interaction. It doesn't require users to execute a dedicated “turn on” command, but rather cleverly integrates interaction into an existing daily action sequence. For instance, it can be used as a bookmark in a book or attached to a refrigerator as a magnet. This allows the user to become the circuit switch itself. The device operates quietly and reliably, providing clear feedback only when needed (e.g., when light levels change). This approach reduces cognitive load and enhances the fluidity of the user experience.



Project Name: Temporary Night Light

Design Statement: Non-Wearable | Material, Technology, Operation

- **Material:**

Copper foil: Used to create circuit contacts on the back and base of the silicone doll.
Battery:
Concealed inside the back to provide power.
LED: Select a small-sized model with the light head exposed from the side or top of the base.
Flat silicone doll: Serves as the surface interaction switch and functional core, with copper foil contacts attached to its back.
Small magnets: Embedded within the silicone doll and base to enable magnetic attachment.

- **Technology:**

Circuit design: A series circuit comprising a battery, LED, and two separate contacts. The circuit closes when the two contacts are bridged.
Switch Mechanism: The silicone doll functions as the switch itself. The copper foil strip on the doll's back serves as the “moving contact,” while the two copper foil pieces on the base act as “fixed contacts.” Placing the doll closes the circuit. Pressing the doll initiates the connection.
Structural Integration: All circuit components are embedded within the flat base, maintaining a clean exterior. The doll attaches magnetically to the base for easy removal and secure holding.

- **Operation:**

Press to Light: To activate the nightlight, simply press the silicone doll bookmark. The circuit contacts on the doll's back connect with the base contacts, automatically illuminating the LED.
Remove to Turn Off: To move or turn off the light, simply detach the doll. This breaks the circuit, causing the light to extinguish immediately. The doll can then function independently as a bookmark.
Multi-Function Magnetization: Thanks to its built-in magnet, the entire device easily attaches to bed frames, refrigerator doors, or other metal surfaces, transforming into a magnetic “fridge magnet.”








Design Statement: Non-Wearable | Design & Visual Inspiration

The core functionality I designed for the “Temporary Night Light” pursues “Gentle Utility.” Rather than chasing flashy tech aesthetics, I sought a warm, clever solution to everyday needs.

Its visual inspiration blends “toy” and “tool” elements. The adorable silicone doll shape dispels the coldness often associated with electronics, making the product feel friendly and personal. Simultaneously, the exposed copper foil circuitry imparts a sense of handmade (DIY) authenticity and comprehensibility. (Alternatively, a plastic casing could be used for concealment.) This design allows users to instantly grasp how it functions.

The light itself is designed as a soft warm yellow—a color temperature ideal for reading and sleep, fostering a tranquil and relaxing atmosphere. The single, non-glaring light source, like a firefly in the night, provides just enough illumination to emphasize its “temporary” and “auxiliary” positioning—not meant to replace primary lighting.

This design language prioritizes functional directness, allowing users to grasp its purpose and operation without instructions. It perfectly addresses the specific, minor pain point of “temporary reading illumination.”







Final models

Project Name:
Glow Headband





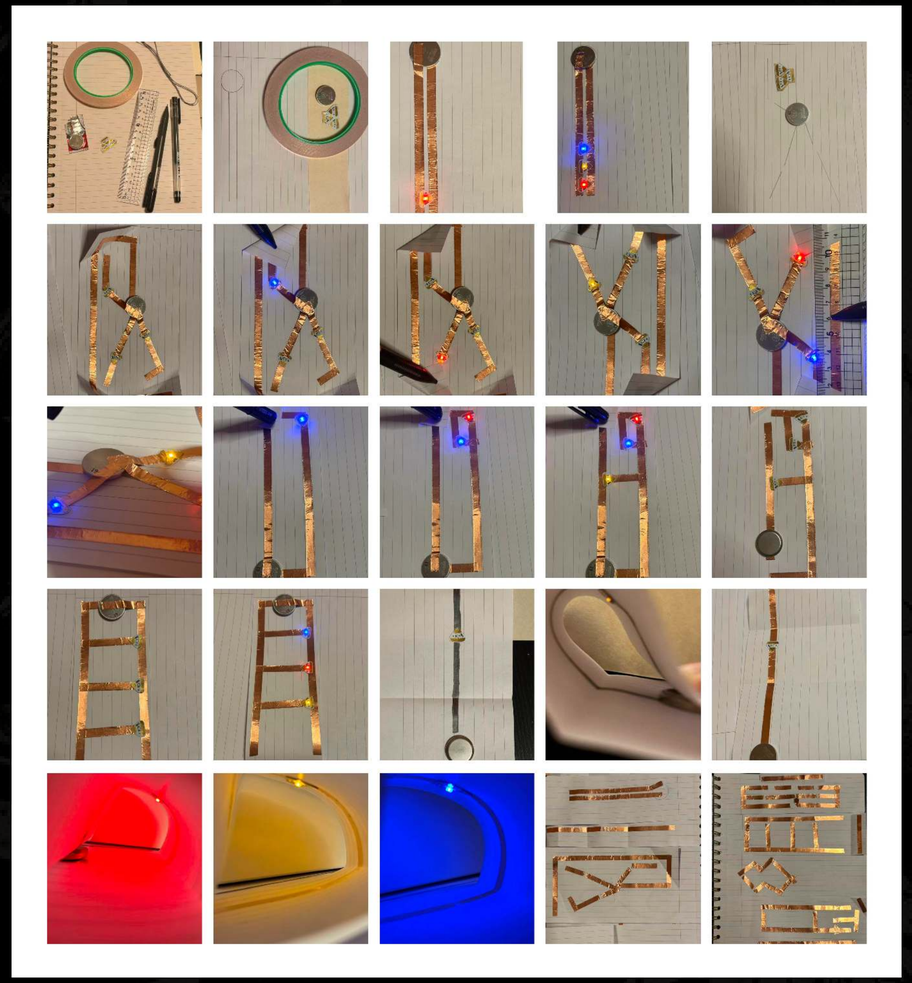
Project Name:
Temporary Night Light





Formstorming

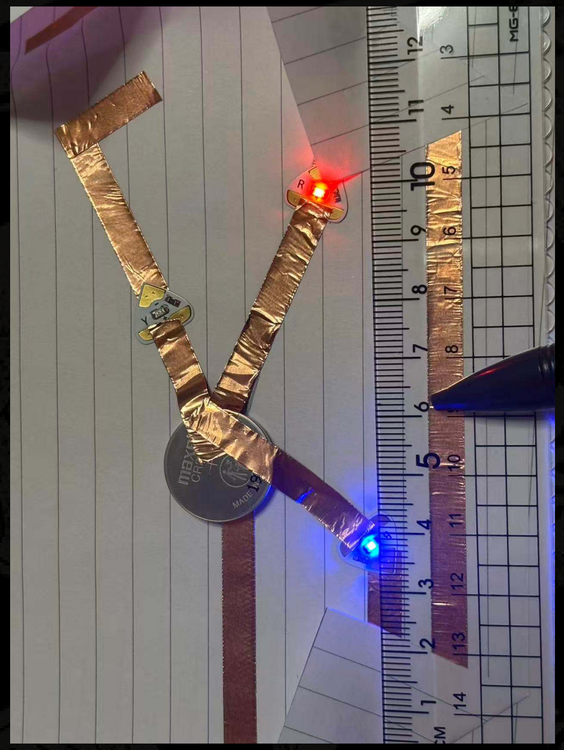
In Activity 1 from Week 2, I constructed a series of interactive circuit prototypes using graphite pencil marks, copper tape, paper, and LEDs. By utilizing materials provided in class, I explored how interactive circuits function and employed various methods to test circuit continuity. This activity profoundly demonstrated to me that materials are not merely physical mediums but also carriers of cultural symbols.

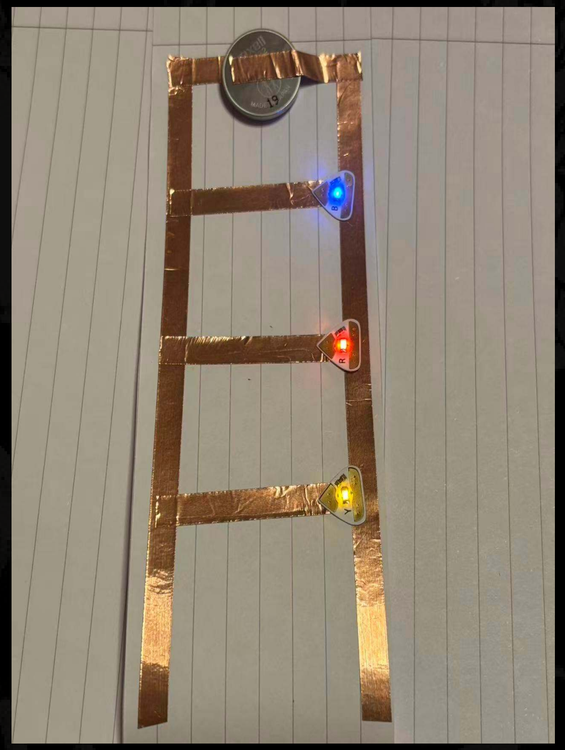


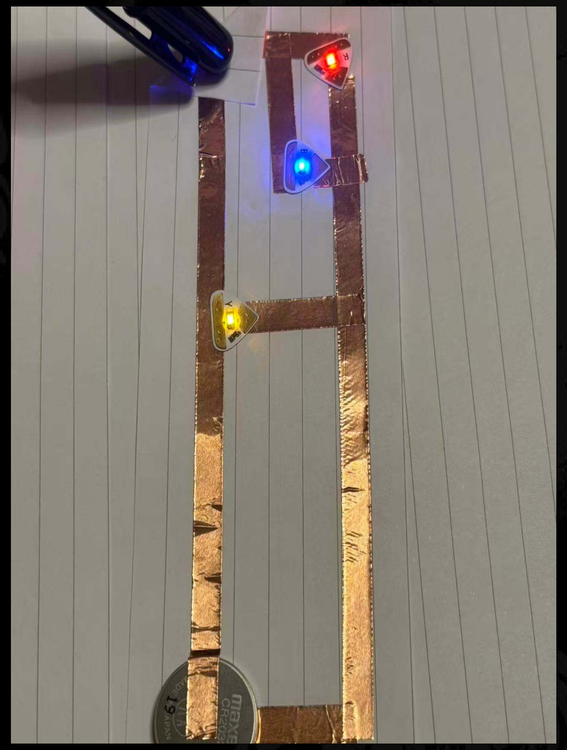
Formstorming

- **Activity 1: Material Exploration and Circuit Fundamentals**

  A series of interactive circuit prototypes were constructed based on the materials provided in the class







Formstorming

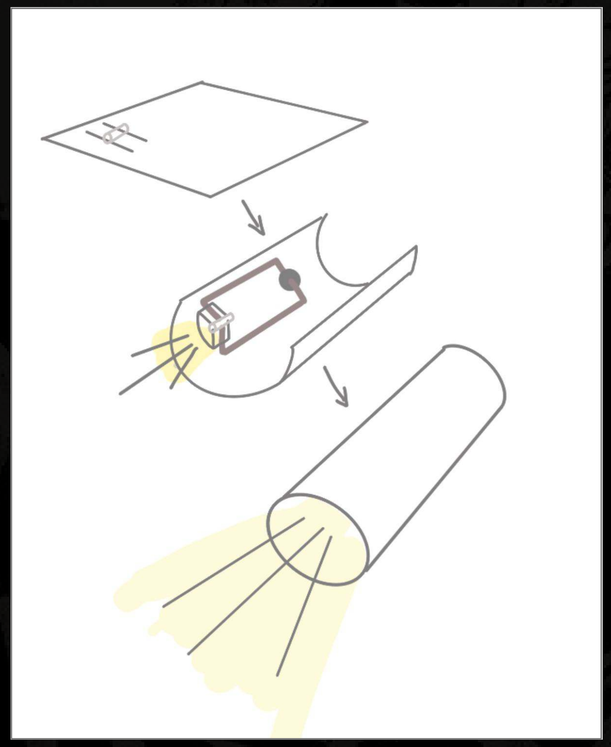
In Activity 2 from Week 3,Building upon the insights from Activity 1, I focused my design on the fundamental concept of minimalist “lighting” and explored how it could be integrated into everyday scenarios. This led to the development of five distinct approaches. I referenced conceptual diagrams and my initial design sketches. Ultimately, I settled on ‘flashlight’ and “bookmark-style night light” as my final design directions.



Formstorming

Simple Flashlight:
This is a Simple Flashlight designed specifically for night time use. A convenient flashlight for emergencies. Its purpose is to provide quick illumination during dark periods through its compact size and ease of use. It can also be worn in multiple ways. This flashlight is made from copper foil, a battery, and an LED light.When you need brief illumination in the dark to find an item or see where you're stepping, simply roll it up gently to turn on the light. Release your hand, and the light turns off.







Reflection: What I Learned

Technical Aspects:
Deepened Material Understanding: Through hands-on practice, I gained a profound grasp of the properties of different conductive materials (such as graphite and copper foil) and their respective advantages and disadvantages in flexible circuit applications.
Circuit Iteration Capability: By completing the activity, I mastered the complete process from conceptual sketches to functional prototypes. Through testing, I identified issues like poor contact and insufficient voltage, enabling me to optimize and iterate the design.
System Integration Mindset: Learned to physically integrate electronic circuits with everyday objects (headbands, plush toys) ensuring functional and form compatibility.

Design Aspects:
Availability-Driven Design: Deeply realized that exceptional design begins with profound insight into an object's inherent properties (Availability), extending functionality from this foundation rather than forcibly layering technology.
Contextual Design Thinking: I recognized that design must be tightly integrated with usage scenarios. The glow-in-the-dark headband was designed for dynamic, social entertainment settings, while the temporary night light was tailored for static, private reading scenarios. Consequently, their interaction logic and visual language were fundamentally different.

Simple is Effective: Both projects validated that the most direct technical solution (a physical switch) often delivers the most reliable and intuitive user experience. Resisting the temptation to over-engineer during the conceptual phase is crucial.

Reflection: Challenges

I think the greatest challenge I encountered stemmed from circuit uncertainties.
Balancing Durability and Comfort in Wearables
In the glow-in-the-dark headband project, maintaining circuit stability during frequent head movements and perspiration posed a significant challenge. In initial prototypes, copper foil fractures occurred after just a few bends.
Solution: Instead of seeking sturdier rigid circuits, I shifted to an optimized “flexible encapsulation” technique. Using high-quality, stretch-resistant copper foil and covering it with a full layer of flexible insulating tape, I “sandwiched” the circuit between them. This significantly enhanced durability.

So: By revisiting the Week 1 tutorial on conductive material properties, I learned to handle copper tape joints with greater precision. This included reinforcing connections with double layers of copper tape. Simultaneously, Week 3's discussion on prototyping helped me embrace the “roughness” of early iterations and focus on testing core functionality.







Reflection: Future Improvements

Future Improvement Directions:
Glow-in-the-Dark Headband: Explore integrating a micro-vibration motor to make the lights flash in rhythm with ambient sounds, enhancing interactivity in music settings. Material-wise, research more invisible, yarn-like conductive fabrics.

Temporary Night Light: Design a series of silicone dolls (various shapes and characters) allowing users to collect “switches” like figurines, boosting product appeal and emotional connection. Alternatively, incorporate a photoresistor for automatic activation in low-light environments.

Final Summary:
Through these two projects, I completed a journey from theory to practice, from identifying problems to creatively solving them. I gained a profound understanding that the core of interaction design lies not in technology itself, but in the relationship between technology and people. Whether it's the “fun” light-up headband or the “useful” temporary night light, the key to success lies in deep observation of user scenarios and the restrained yet precise application of technical means.

Citations

- Course documents:

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WEEK 3: No link. It's from the courseware.

- IMAGE:

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THANK YOU